

EXECUTIVE BRIEFING

Predicting Employee Attrition & Data-Driven Retention Insights

Executive Summary: Retaining high-performing employees is a critical driver of organizational stability and productivity. Using advanced predictive analytics on our historical workforce data, we developed an early-warning system that identifies employees likely to leave the company. The model is highly effective, successfully capturing **66% of all actual departures** in advance with an overall predictive accuracy of **80%**. This report translates these statistical findings into clear, actionable insights and concrete recommendations for our HR retention strategy.

Key Drivers of Employee Attrition

Our analysis reveals three primary factors that strongly predict whether an employee will leave:

- **Excessive Overtime:** This is the single strongest driver of employee departures. Employees working overtime are **3.5 times more likely to leave** than their peers. This points directly to burnout and work-life imbalance as a primary exit route.
- **Frequent Business Travel:** Employees who travel frequently for work experience an attrition rate of **24.9%**, compared to just 11.5% for those who rarely travel or do not travel. The strain of constant travel significantly degrades employee retention.
- **Low Salary and Compensation:** Employees who left the company earned a median salary of **\$3,202 per month**, which is **38% lower** than the median salary of those who stayed (\$5,204). While salary is not the sole factor, low pay remains a massive baseline vulnerability.

Who is Most at Risk? (Target Segments for Retention)

HR should prioritize retention efforts on the following departments and job roles where attrition is highly concentrated:

1. **Sales Representatives (39.8% exit rate) and Laboratory Technicians (23.9% exit rate):** These front-line roles have exit rates far exceeding the overall company average of 16.1%.
2. **New Hires (Tenure under 2 years):** Employees are highly vulnerable early in their tenure. Attrition among employees in their first year is **34.5%** and **36.4%** for those with less than a year, dropping to 10.7% by year five. This suggests a critical gap in early onboarding and integration.
3. **Work-Life Balance Score of 1 (31.3% exit rate):** Employees who rate their work-life balance as poor leave at more than double the rate of those rating it as good or excellent.

Strategic Recommendations for HR

Based on these insights, we recommend implementing the following targeted initiatives:

1. **Implement an 'Overtime Cap & Audit' Policy:** Establish strict thresholds for overtime hours in high-risk areas (like Laboratory Technicians). HR should conduct monthly audits of teams working high overtime and implement compensatory time-off or team load-balancing to prevent burnout.
2. **Revamp Early Career Integration (Mentorship Program):** Launch a 12-month structured mentorship program specifically targeting new hires in their first year. Proactive HR check-ins at 30, 90, and 180 days will help identify early frustrations before they lead to resignation.
3. **Targeted Compensation Adjustments:** Conduct a salary equity review for Sales Representatives and Lab Technicians. Ensure base pay aligns with market standards, and link incentives to tenure and performance to make early-career compensation more competitive.

Model Scope and Limitations

Important Usage Context: While this predictive model is highly effective, it has limitations. It is built on historical internal data and does not capture external market forces (such as competitor hiring campaigns or industry salary spikes) or personal life changes (such as relocation or family needs). The model should be used as a *supportive tool* to guide HR's outreach, not as a replacement for direct human conversation. We recommend retraining the model quarterly with updated workforce data to ensure its predictions remain accurate as company policies evolve.